DIGITALEDUCATIONREVIEW

Implementing a Recommendation System to Predict Course Selection for Higher Education

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ABSTRACT

In order to encourage the students to follow an inter-disciplinary education system where they can study multiple academic disciplines, the New Education Policy offers them a variety of elective courses. Greater flexibility promotes all-around development and can cater to a diverse set of interests, but it poses a serious issue as well: students do not know how to choose electives that align with their academic abilities, career aspirations, and personal interests. A content-based filtering and collaborative filtering technique-based recommendation system is recommended in this research to solve this issue and assist students in making informed elective choices.

Personalized course suggestions are provided by the strategy on the basis of data like students' past performances, specific areas of interest, and courses available. All this data is examined by the algorithm and recommends electives that best fit the student's goals and ability level. It seeks to simplify the voluntary decision-making process, reduce decision anxiety, and improve academic performance and engagement.

The efficacy of the concept is determined by a sequence of pre- and post-tests of educational achievements and student satisfaction after the implementation of the recommendation system. Moreover, the study provides an in-situ and scalable solution to one of the biggest problems of modern multidisciplinary education.

KEYWORDS: Higher Education, Content-Based filtering, Collaborative Filtering, Hybrid, machine Learning

1. INTRODUCTION

The traditional Indian educational strongly emphasis on areas like language, science, and arithmetic, this system has adhered to a restricted, formalized curriculum. This educational system has been effective in providing students with the foundational knowledge they need to thrive academically, but it falls short in its more crucial function of encouraging specialized learning and intellectual inquiry. This rigidity prevents students from learning a wider range of things that could be more directly tied to their personal interests and professional aspirations as they advance through the educational system. Because of this, the majority of students only have a superficial understanding of their academic area when they graduate (Neha et al., 2023).

To cater to the changing needs of the world economy of the 21st century, educational systems all over the world—India included—have undergone a revolutionary transformation in the recent past. One such initiative in this regard is the National Education Policy (NEP) 2020, an evolutionary transformation in India's educational system. At each level of education, the NEP provides a more multidisciplinary, adaptable, and integrated education. The final aim of higher education is to enable students to opt for a set of disciplines, including micro-level and transdisciplinary ones not offered or underrepresented earlier (Kruthika G. V. et al.,2023).

The institutionalization of such a compromising curriculum is marred by a host of new problems despite its reform agenda. The sheer multitude and richness of choice facing students nowadays is perhaps the most pressing among these. When it comes to choosing their subject areas, students are swamped by the multitude of elective, specialist, and role-specific courses on offer (Christine Lahoud et al., 2022). Variable degrees, curriculum demands, and the uncertainty of potential career directions further complicate this. So many options may be particularly overwhelming for those who are still unclear about their academic or career aspirations.

Students are subjected to poor habits such as poor decision-making in such a scenario. Instead of giving an honest evaluation of their abilities and interests, they will select electives based on social pressure, popularity, or mere guessing. As a result, they become disinterested, receive low grades, and are unhappy with their college experience throughout.

Schools and institutions are now employing technology, i.e., Recommendation Systems (RS), to reduce such issues and ease the decision-making process. A system that can forecast human interests based upon an analysis of history data, behavioural patterns, and similarity of items is referred to as a RS. To make accurate and personalized recommendations, it typically employs hybrid methods, Content-Based Filtering, or Collaborative Filtering. Through continuous learning and adaptation, the systems enhance the saliency of results and provide efficient information retrieval through modelling user-item interactions based on machine learning and data mining. Recommendation Systems, which were initially developed and widely implemented in fields such as e-commerce and web entertainment, have proven to have a stunning capability of making tough decisions easy by recommending something on the basis of user history (Joy Dhar and Asoke Kumar Jodder, 2020; Kruthika G. V. et al.,2023). The same algorithms can be applied to education to assist students in selecting optional courses that are appropriate with their planned objectives, previous performance, and field of study.

A course recommendation system design for higher learning institutions is suggested by this work. The system combines two different recommendation strategies: collaborative filtering and content-based filtering. Content-based filtering, on the basis of the student's past academic record and areas of interest, recommends courses that are very similar to the ones they have attended or are keen on. Collaborative Filtering, in contrast, uses data from other students with similar profiles or interests in an effort to identify patterns and recommend courses to them (Mehdi Elahi et al. 2022).

The system tries to provide every student with a uniquely personalized set of electives through these two mechanisms. These electives will be individually chosen to align with the student's long-term academic and professional aspirations as well as be suitable to his or her academic ability and interest. The three primary goals of the system are to reduce the cognitive burden of course selection, make evidence-based decisions, and maximize academic performance (Luxiao Zhu and Ben Wang,2022). By facilitating students to make meaningful, personal, and informed education decisions in an adaptive academic environment, it hopes to make the NEP 2020 vision more achievable.

To minimize the cognitive burden of the students which he/she experiences during course selection, to support research-based academic planning through the integration of analytics, historical performance data, and predictive modelling and to enhance overall academic outcomes by recommending course pathways are the three objectives that research is focusing and trying to propose a model fulfilling these.

2. REVIEWING AND EXAMINING RECOMMENDATION SYSTEM

There are four types of RS available from which in this research two were used in the implementation of this paper: Content-Based Filtering and Collaborative Filtering. A short overview of the applied recommender system is given below.

2.1 Content-based filtering

Content-Based Filtering is a popular recommendation method that recommends items to users on the basis of past choices and the characteristics of the items. In educational technology, this method can be used very effectively to offer personalization in course suggestions by evaluating user information and course details (– Adhvaryu and Dr. Disha Parekh, 2024; F. Furtado and A. Singh, 2020). The core process is about aligning the features of courses that were previously chosen or successful with those currently available, thus making recommendations that cater to the personal academic interests, aptitude, and profile of the individual user (Tejashri et al., 2024).

This method is based on specific user characteristics—explicit, like declared majors and expressed interests, and implicit, like academic history. The completer and more accurate the information that the user submits, the better the recommendations will be. It is this degree of personalization that makes possible a focused educational experience. For example, a student who has proved to be highly proficient in maths and interested in computer science can be advised to take electives such as "Machine Learning" or "Data Structures," which will be most compatible with his or her academic background (Aamir Fareed et al., 2023; Maria Cora et al. 2021).

One of the defining strengths of Content-Based Filtering is that it can make very personalized recommendations without considering other users' behaviour or preferences. This freedom means that the system is not susceptible to popularity bias but rather concentrates exclusively on the applicability of course content to the specific user. The method is not perfect, however. It relies quite greatly upon the quality and availability of student and course metadata. Poor or incomplete data will, in turn, hamper the system's capacity to make useful recommendations (N D Lynn, et al., 2021).

In addition, Content-Based Filtering also has the "cold start problem." The problem is encountered when the system is presented with new users or new courses for which there is not a large amount of historical data. With limited data, the system cannot make as precise a prediction, thus initially losing its effectiveness. Another shortcoming is the "randomness problem," whereby the system keeps suggesting similar kinds of courses, and in doing so, might limit the student's range of exposure and fail to promote academic discovery or cross-disciplinary learning (Murtala et al., 2020).

A number of algorithms are typically used in carrying out Content-Based Filtering. These are (Jing Li and Zhou Ye, 2020; Yash Kadam et al., 2023):

Term Frequency (TF): Quantifies the number of times a term occurs within a document, expressing its relevance in that particular context.

Inverse Document Frequency (IDF): Emphasizes words that are distinctive or unique in a dataset by discouraging frequently used words, thus enhancing discrimination between documents.

Cosine Similarity: Computes the cosine of the angle between two non-zero vectors, essentially measuring how much two course profiles are similar in terms of content.

K-Nearest Neighbours (KNN): A supervised learning model that compares similarity between data points and is therefore well suited to finding closely related courses.

Decision Trees: Models that are illustrated in a graphical manner with multiple decision paths and outcomes, which can help classify courses according to user preferences and characteristics.

In school environments, these computing devices enable Content-Based Filtering to function with significant complexity, making recommendations that catalyse student motivation and academic achievement. With some constraints, this method possesses significant promise in personalizing the learning experience and assisting students with selecting optimal courses.

2.2 Collaborative filtering

Collaborative Filtering is a robust recommendation method that recommends items to users by relying on the actions and interests of similar users (Juan S. et al., 2023). Collaborative Filtering differs from Content-Based Filtering, which is attribute-based, since it is pattern-based in user behaviour. It works on the assumption that users with common taste in the past are likely to have common tastes in the future too (K. Bhumichitr and S. Channarukul, 2017).

In the case of course recommendation systems, Collaborative Filtering suggests students with comparable academic profiles, interests, or course history and recommends popular or highly rated electives among their peers (Shrooq Algarni and Fredrick Sheldon, 2023). For example, if some students with comparable performance measures and data science interests have taken a certain machine learning course, the system might suggest that course to a prospective student with a comparable profile. This peer-driven methodology allows the system to reveal hidden patterns that aren't directly visible through conventional data attributes by itself (D. Phani Kumar et al., 2023).

The Collaborative Filtering diversify a student's course choice by taking advantage of other users with the same course choices, it can recommend courses students might not have been acquainted with otherwise, extending their familiarity with academic material beyond their former choice or declared interests (Xingjuan C. et al., 2020). It is particularly valuable in enabling transdisciplinary learning and discovery. There are, nevertheless, certain extremely critical limitations to Collaborative Filtering. It is plagued by the notorious "cold start problem" in which the system cannot make good recommendations since it has insufficient information about new users or recently added courses (Murtala et al., 2020).

To overcome these obstacles, certain algorithms in machine learning are applied. They are Bayesian networks, matrix factorization algorithms, user-based k-nearest neighbours (KNN), and clustering models. These algorithms in machine learning help identify and strengthen the user-item relationship even under incomplete data (Akshita Gupta and Anand Sharma, 2020).

Overall, Collaborative Filtering provides a great deal of value to educational recommendation systems through peer-informed course choice and the discovery of latent trends in student behaviour. While there are some limitations to its application, its potential to provide personalized and exploratory recommendations makes it a valuable element of adaptive learning technologies (Qian Zhan et al., 2020).

2.3 Hybrid recommendation system

By combining an assortment of recommendation methods, a hybrid recommendation system can enhance precision, overcome the drawbacks of individual approaches, and generate more precise and targeted recommendations.

Content-based filtering, based on item features and user tastes, and collaborative filtering, based on user-item interactions, are the two general categories into which most traditional recommendation systems have traditionally been classified. Each is good at something, but each has its limitations. To eliminate these, hybrid systems use a variety of algorithms in a complementary manner (Amane M. A, 2022).

There are several hybridization schemes are possible, such as switching, feature combination, weighted, and cascade. Multiple recommenders are combined employing learned or fixed weights in a weighted hybrid. Switching approaches are employed to choose a recommendation algorithm over another dynamically according to certain criteria or context of data. Feature combination generates one model by combining information from multiple sources, and cascading enhances by employing recommenders in a sequential process (Luxiao Zhu and Ben Wang,2022; Akshita Gupta and Anand Sharma, 2020).

The capacity of hybrid recommendation systems leverages the strengths of various approaches while avoiding their weaknesses has fostered an enormous amount of research into the systems. Personalization of recommendations for certain applications and data sets can boost recommendation quality across a variety of categories. Adaptive weights and feature learning from complex, high-dimensional data have become possible with machine learning and deep learning, which have further empowered hybrid systems.

When there are recommendation tasks, hybrid recommendation systems are a robust and flexible solution. Next-generation intelligent recommendation engines place a high degree of focus on them as they enhance efficiency, flexibility, and customer satisfaction by integrating various methods (Qian Zhan et al., 2020; Amane M. A, 2022)

3. IMPLEMENTATION STEP

The implementation phase assists in the bridging of ideas at the theoretical level to real-world applications. A preferred framework is translated into possible actions that assist others understand how the research might be carried out in practical situations. It further helps check the functionality, accuracy, and impact of research against pre-set goals or standards.

This study uses recommender system methods like Content-Based methods and Collaborative Filtering separately in a bid to advise students in choosing their electives. Three tables constitute the implementation dataset to be used in an effort to implement the algorithm implementation: In the first one, students' professional goals will be briefly described; in the second, the course description; and in the third, the course selection and rating. Here Synthetic data are used in all these below three tables.

user_df(user_id,major,career_goal)

course_df(course_id,title,department,difficulty_level)

interaction_df(user_id,course_id,rating)

The information provided by User_df will include the student's field of study and career goals. For Example, user_id=1, major=Computer Science and career_goal = Software Engineer.

Course_df will contain basic information such as the course name, department from which students can take it, and degree of difficulty. For Example, course_id = 101, title = Introduction to Python, Department = Computer

Science and rating = Beginner User_id and course_idare used in both datasets to identify users and courses, respectively.

Interaction_df will give basic information like which students take which course and rating given by them. For Example, user_id = 1, course_id = 101 and rating =4.

In the Content-Based Filtering approach, the algorithm begins by importing a course description and student dataset. A Term Frequency–Inverse Document Frequency (TF-IDF) matrix is computed for every course offered to represent the keyword significance of a course in numeric form. The second is pre-processing student information. If the student's preferences or interests are already in TF-IDF matrix form, then the algorithm proceeds; otherwise, the student information is transformed into an equivalent TF-IDF representation with the same vocabulary as that used for course data. Once having both matrices available, cosine similarity is employed to compute similarity between the student profile and specific courses. Similarity values are sorted in decreasing order, and the top N courses with highest similarity values are recommended to the student. The approach ensures that recommendations are in accordance with the stated interests and studies of the student.

The flowchart and code snippet of Content-Based Filtering are as follows:

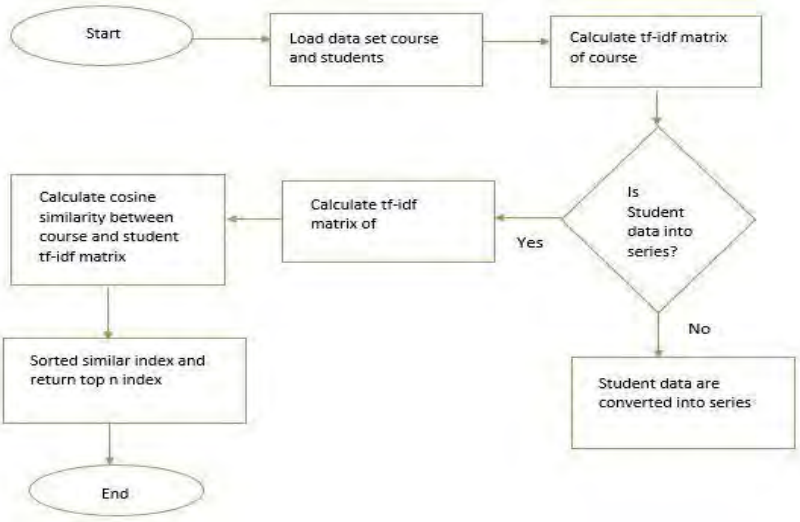


Figure 1 Flow chart of Content-Based Filtering algorithm

```
interaction_df['features'] = interaction_df['user_id'].astype(str) + ' ' + interaction_df['course_id'].astype(str) + ' '
+ interaction_df['rating'].astype(str) # Changed course_df to interaction_df
print(interaction_df['features'])
tfidf_vectorizer = TfidfVectorizer()
tfidf_matrix = tfidf_vectorizer.fit_transform(interaction_df['features'])
print(tfidf_matrix)
def get_content_based_recommendations(user_profile, top_n=5):
     if not isinstance(user_profile, pd.Series):
        user_profile = pd.Series(user_profile)
    user_profile = user_profile.astype(str)
    user_profile_features = ' '.join(user_profile.values) # Remove the parentheses after .values
    user_profile_vector = tfidf_vectorizer.transform([user_profile_features])
    cosine_similarities = cosine_similarity(user_profile_vector, tfidf_matrix).flatten()
    recommended_indices = cosine_similarities.argsort()[-top_n:][::-1]
    recommended_indices = [idx for idx in recommended_indices if idx < len(course_df)]
    return course_df.iloc[recommended_indices]
```

Figure 2 Snippet code of Content Base Filtering

The Collaborative Filtering approach is based on learning from patterns of student-course interactions. Loading a user interaction dataset with various courses, for example, ratings from 1 to 5, is the initial step. Synthetic ratings are employed to substitute actual ratings in case they are missing. The system then identifies all unique courses in the dataset and applies the K-Nearest Neighbours (KNN) algorithm to investigate user similarities. Based on this strategy, the system is able to predict users who interacted with similar courses in a similar manner. Similar users' interests are then examined in order to determine what courses are usually selected or rated highly. Sorting these results and bringing back the best N courses as personalized recommendations to the target student is the final process.

The flowchart and code snippet of Collaborative Filtering are as follows:

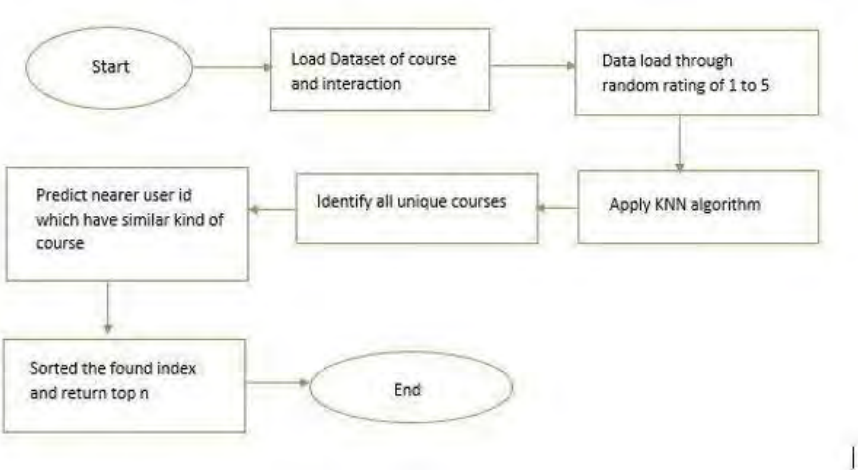


Figure 3: Flow chart of collaborative filtering algorithm

```
reader = Reader(rating_scale=(1, 5))
data = Dataset.load_from_df(interaction_df[['user_id', 'course_id', 'rating']], reader)
trainset, testset = train_test_split(data, test_size=0.25)
model = SVD()
model.fit(trainset)

def get_collaborative_recommendations(user_id, top_n=5):
    courses = course_df['course_id'].unique()
    predictions = [model.predict(user_id, course) for course in courses]
    top_predictions = sorted(predictions, key=lambda x: x.est, reverse=True)[:top_n]
    top_courses = [pred.iid for pred in top_predictions]
    return course_df[course_df['course_id'].isin(top_courses)]
```

Figure 3: Snippet code of Collaborative Filtering

By integrating these two methods of filtering, the system provides a solid and data-driven solution to the course selection problem. Content-Based addresses relevance in the context of course content and student interest, and Collaborative Filtering identifies the common tastes among group peers. In addition to relieving the decision-making process for students with an enormous list of courses in front of them, this two-layered approach allows students to make informed decisions that are consistent with their long-term career goals and academic abilities.

4. RESULTS

The execution of Collaborative Filtering and Content-Based algorithms, the recommendation system is then tested using a confusion matrix and metrics like Accuracy and Precision and test cases to check whether the system is working and can be deployed in the real world.

The following Result illustrates the confusion matrices of the Collaborative Filtering and Content-Based models of recommendation, with precise assessment of their classification performance. The left matrix is the Collaborative Filtering model, and the matrix on the right illustrates the results of the Content-Based Filtering model.

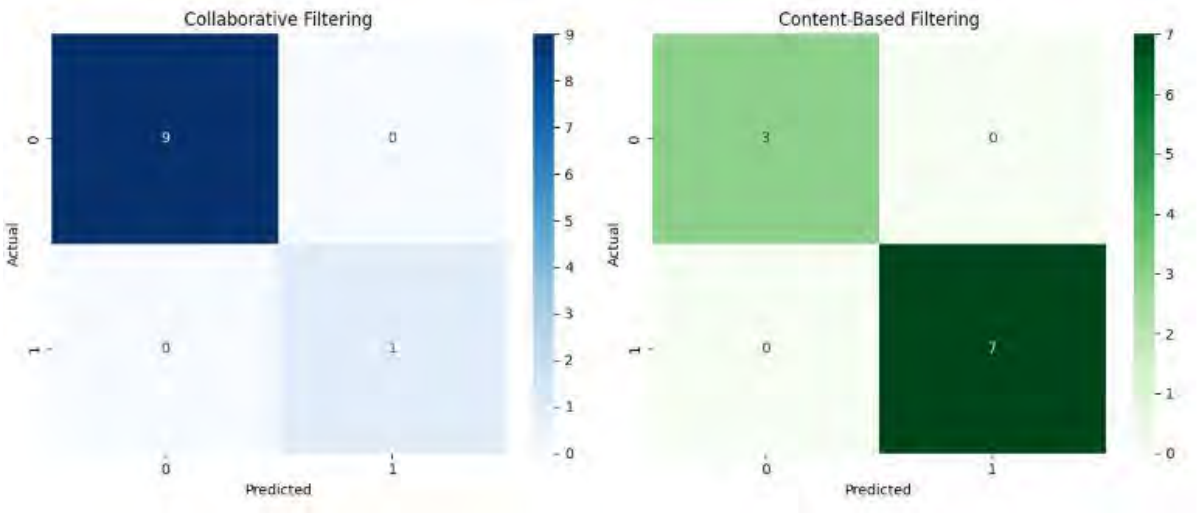


Figure 4: Confusion Matrix of the Result

For both models, the confusion matrices demonstrate perfect classification accuracy on the provided test cases. Specifically:

In the Collaborative Filtering model, all 9 negative instances (i.e., users who were not expected to be interested in the recommended course) were correctly classified as negative (True Negatives), and the single positive instance (i.e., a user expected to engage with the course) was correctly classified as positive (True Positive). No false positives or false negatives were observed.

Similarly, the Content-Based Filtering model correctly classified all 3 negative instances and all 7 positive instances, with zero misclassifications.

In addition to visual analysis via the confusion matrix accuracy, and Precision are computed to quantify the predictive performance of both recommendation systems. These metrics is defining as

$$\text{Accuracy} = (\text{TP} + \text{TN})/(\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

and $\text{Precision} = \text{TP}/(\text{TP} + \text{FP})$

Where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives

In Collaborative Filtering TP=1, TN=9, FP=0, and FN=0, so accuracy of the model is 1 and precision of the model is also 1. These shows Collaborative Filtering achieved perfect performance and it correctly classifying all the instances in the test set without any false positives or false negatives.

In Content-Based Filtering TP=7, TN=3, FP=0, and FN=0, so accuracy of the model is 1 and precision of the model is also 1. Similarly, the content-based filtering system demonstrated flawless classification on the provided test data.

The test cases below will explain the result analysis of Content Bases and Collaborative Filtering technique. Here ten cases are used for the domain to determine the outcome.

Test Cases of Content Base Recommendation System											
Sr. No	Input Data				Expected Output			Actual Output			Result
	Student Id	Course Id	Title	Difficulty Level	CourseId	Title	Difficulty Level	Course Id	Title	Difficulty Level	
1	1	101	Introduction to Python	Beginner	104	Introduction to R	Intermediate	104	Introduction to R	Intermediate	Pass
2	5	103	Trigonometry	Advanced	111	Complex Analysis	Advanced	111	Complex Analysis	Advanced	Pass
3	7	104	Introduction to R	Intermediate	109/110	Machine Learning Concept/Deep Learning	Advanced	104	Introduction to R	Intermediate	Fail
4	9	105	Introduction to Physics	Beginner	106	Electromagnetism	Intermediate	105	Introduction to Physics	Beginner	Fail
5	12	106	Electromagnetism	Intermediate	107/112	Nuclear Pyhsics / General Relativity	Advanced	112	General Relativity	Advanced	Pass
6	18	102	Geometry	Intermediate	103	Trigonometry	Advanced	103	Trigonometry	Advanced	Pass
7	24	107	Nuclear Pyhsics	Advanced	112	General Relativity	Advanced	112	General Relativity	Advanced	Pass
8	26	110	Deep Learning	Advanced	109	Machine Learning Concept	Advanced	110	Deep Learning	Advanced	Fail
9	28	111	Complex Analysis	Advanced	103	Trigonometry	Advanced	103	Trigonometry	Advanced	Pass
10		102	Geometry	Intermediate							
	30	111	Complex Analysis	Advanced	103	Trigonometry	Advanced	103	Trigonometry	Advanced	Pass

Table 1 Test Cases of Content Base Recommendation System

This paper proposes a Content-Based recommended system that recommends courses to students according to their interests and the characteristics of the course. But it gives wrong results in many situations, for example

1. Because they rely on course components given, students with certain interests in CBRS may not find a suitable RS easily.
2. The specified interest of the student is too narrow; hence, the algorithm can recommend only similar items in different forms. It is missing out on Content that might be more interesting or diverse.

Test Cases of Collabrative Filtering Recommendation System											
Sr. No	Input Data				Expected Output			Actual Output			Result
	Student Id	Course Id	Title	Difficulty Level	CourseId	Title	Difficulty Level	Course Id	Title	Difficulty Level	
1	1	101	Introduction to Python	Beginner	104	Introduction to R	Intermediate	105	Introduction to Physic	Beginner	Fail
2	5	103	Trigonometry	Advanced	111	Complex Analysis	Advanced	102	Geometry	Intermediate	Fail
3	7	104	Introduction to R	Intermediate	109/110	Machine Learning Concept/Deep Learning	Advanced	108	Introduction to Algebra	Beginner	Fail
4	9	105	Introduction to Physics	Beginner	106	Electromagnetism	Intermediate	105	Introduction to Physics	Beginner	Fail
5	12	106	Electromagnetism	Intermediate	107/112	Nuclear Pyhsics / General Relativity	Advanced	107	Nuclear Physics	Advanced	Pass
6	18	102	Geometry	Intermediate	103	Trigonometry	Advanced	107	Nuclear Physics	Advanced	Fail
7	22	109	Machine Learning Concept	Advanced	110	Deep Learning	Advanced	105	Introduction to Physics	Beginner	Fail
8	24	107	Nuclear Pyhsics	Advanced	112	General Relativity	Advanced	105	Introduction to Physics	Beginner	Fail
9	28	111	Complex Analysis	Advanced	103	Trigonometry	Advanced	108	Introduction to Algebra	Beginner	Fail
10		102	Geometry	Intermediate							
	30	111	Complex Analysis	Advanced	103	Trigonometry	Advanced	105	Introduction to Physics	Beginner	Fail

Table 2: Test Cases of Collaborative Filtering Recommendation System

The Collaborative recommended system, based on the interests and actions of similar users, suggests a course; but CBRS supplied wrong result since

1. Students have interacted with only a few of the available items in a course; sparse interactions between students may produce less accurate results from RS.
2. Because the system relies so heavily on student similarity, it can easily suggest the wrong course based on superficial similarities. For example, instead of looking at more meaningful decisions, the algorithm might suggest a course based on the few shared interests of two students.
3. Collaborative filtering can lead to a less diverse set of recommendations, often highlighting courses that are like ones a student has liked before. This method can never expose users to unexpected results.

5. DISCUSSION

The objective of this study is fulfilled by giving rational and personalized course recommendations. The study is used to find the reducing cognitive load which enables students to engage more effectively with optimized learning pathways. The study further reveals that this reduction in decision-making demand establishes the infrastructure necessary for adaptive, student-centric learning at scale. These findings confirm that the system not only fulfils the stated objectives but also demonstrates their practical relevance by transforming policy goals into actionable institutional practices. By aligning with NEP 2020's vision of flexible, choice-based curricula, the system demonstrates its capacity to bridge the gap between policy aspiration and institutional implementation. This alignment ensures that personalized academic guidance is accessible to all learners, thereby addressing equity concerns and preventing such support from remaining a privilege reserved for a select few. Importantly, the findings highlight the system's broader relevance by not only just fulfilling the stated objectives but also by illustrating how the technological infrastructure can translate national education policy into scalable, student-centric practices that reshape higher education.

References	Algorithm/Approach	Similar Approach with Current Study	Different Approach in Current Study	Current Study Contribution
María Cora Urdaneta-Ponte et al., Adhvaryu et.al, Lynn et.al, Algarni et.al	Systematic Review	Suggest hybrid approach for recommendation	Review perspective	Theoretical findings of the review are supported by practical implementation

Neha, Sharma & Kaur	Represent the survey of Content Based Filtering and Collaborative Filtering	Suggest of using hybrid approach to solve cold star problem	Focuses on survey not on practical implementation	Practical implementation for course recommendation
Kruthika & Divakar	Content based Filtering	Implement content-based filtering	Focus only on content-based filtering	Implement both content base and collaborative filtering
Fareed et al.	Used Collaborative Filtering Technique	Pattern Recognition through Collaborative	Integrate addition attribute using social network	Shows improved pattern recognition using improved meta data
Li & Ye	Used Collaborative Filtering Technique	Shows problem like sparsity and cold star	Used only collaborative filtering	Perform practical implementation of Content Based and Collaborative Filtering along with the problem discussion and solution
Mondal et al.	Used Collaborative Filtering Technique	Creating a personalized recommendation using academic performance	Focus on grade, use metric specific approach and used only collaborative filtering	Use comprehensive meta data for academic marks

Table 3: A comparison between the study conducted till date and the current study carried out.

This research fundamentally extends traditional recommendation system frameworks by moving beyond entertainment or commercial platforms. This study reconceptualize user similarity in education to incorporate learning pathway and to develop academic progression plan. This domain specific study represents that educational recommendation require fundamentally different approach in recommendation.

The major practical challenges associated with implementing recommendation systems in higher education are not taken care of in this work. For instance, the difficulty of collecting a student's data from educational institutions combines the complexity of organizing heterogeneous data within different pedagogical approaches, unlike customer transaction data on commercial platforms, which is more readily available and often shows uniformity.

The application of recommendation systems to higher education fundamentally challenges existing theories of learning by demanding paradigmatic shifts away from standardized to personalized learning paradigms. The practical implementation of recommendation system in higher education give impact to multiple stakeholders within the higher education. Firstly, Academic Advisors which able to give more informed guidance to the students by identifying courses aligned with student preferences and academic performance. Secondly, curricular designer and institutional administrators, recommendation system help them in finding patterns in student course selection which help them in introducing new program, scheduling of the courses, it also helps them by finding which course cluster together in student preferences that maximize student enrolment and resource allocation. It also helps students, in identifying courses by giving personalized guidance.

Additionally, our study represents that student educational performance directly correlated with recommendations accuracy which is measured with commercial context. If result give you wrong suggestion it directly reflect to student educational performance. To solve this problem, in this study author give suggestion of developing a hybrid algorithm by combining Content-based Filtering and Collaborative Filtering to guide students in choosing their course. When Content-based Filtering effectively make use of Course metadata to assist new students it has limited suggestion scope and is unable to adapt changes in students' interest. On the other hand, Collaborative Filtering is extremely good at identifying user trends but facing sparse interactions matrices, which make it difficult to find patterns who are new for the system. Combination of these two algorithms will help to overcome the problem which it is facing while implementing alone.

6. CONCLUSION AND FUTURE WORK

Through Content-Based and Collaborative RS, this paper tries to help students choose optional courses. The student's different information, such as their domain, the previously chosen courses, and their rating, is inputted into the algorithm. The result of the experiment proves that the algorithm cannot solve the problem in all situations and that the use of it makes the participants worse because it creates more confusion.

This paper depicts the implementation of two types of recommender systems. Though the recommender systems used are shown with various reasons of failure, it was evaluated that using a hybrid approach should give a better understanding of the recommendations. In future, the target is to implement a hybrid approach using both the recommender systems. The study of different papers targeting hybrid approaches has already been conducted and later the implementation by combining the different approaches of RS will be carried out.

This research has fulfilled its long-term objectives mentioned in the introduction. The first objective is satisfied by offering a structured, guided decision-making process which aids students to reduce burden while course selection. As the model implemented in research ensures the recommendations are grounded in reliable insights enables to fulfil second objective that supports evidence based academic planning. Both the solutions that fulfils two objectives ultimately fulfils the third objective of the research which enhances the overall academic outcomes by identifying and suggesting various course pathways.

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IMPLEMENTACIÓ D'UN SISTEMA DE RECOMANACIÓ PER PREDIR L'ELECCIÓ D'ASSIGNATURES EN L'EDUCACIÓ SUPERIOR

Per tal d'animar els estudiants a seguir un sistema educatiu interdisciplinari on puguin estudiar múltiples disciplines acadèmiques, la Nova Política Educativa els ofereix una varietat d'assignatures optatives. Una major flexibilitat promou un desenvolupament integral i pot atendre un ventall divers d'interessos, però també planteja un problema seriós: els estudiants no saben com triar assignatures optatives que s'ajustin a les seves capacitats acadèmiques, aspiracions professionals i interessos personals. En aquesta investigació es recomana un sistema de recomanació basat en tècniques de filtratge per contingut i filtratge col·laboratiu per resoldre aquest problema i ajudar els estudiants a prendre decisions informades sobre les assignatures optatives.

L'estratègia proporciona suggeriments de cursos personalitzats basats en dades com el rendiment anterior de l'estudiant, les àrees d'interès específiques i els cursos disponibles. Totes aquestes dades són analitzades per l'algorisme, que recomana les assignatures optatives que millor s'ajusten als objectius i al nivell de capacitat de l'estudiant. Pretén simplificar el procés de presa de decisions voluntària, reduir l'ansietat per prendre decisions i millorar el rendiment acadèmic i la participació.

L'eficàcia del concepte es determina mitjançant una sèrie de proves pre i post sobre els assoliments educatius i la satisfacció de l'estudiant després de la implementació del sistema de recomanació. A més, l'estudi proporciona una solució in situ i escalable a un dels problemes més grans de l'educació multidisciplinària moderna.

PARAULES CLAU: Educació Superior, Filtratge basat en contingut, Filtratge col·laboratiu, Híbrid, aprenentatge automàtic

IMPLEMENTACIÓN DE UN SISTEMA DE RECOMENDACIONES PARA PREDECIR LA SELECCIÓN DE CURSOS EN LA EDUCACIÓN SUPERIOR

Con el fin de animar a los estudiantes a seguir un sistema educativo interdisciplinario en el que puedan estudiar múltiples disciplinas académicas, la Nueva Política Educativa les ofrece una variedad de cursos optativos. Una mayor flexibilidad promueve el desarrollo integral y puede satisfacer una amplia gama de intereses, pero también plantea un problema grave: los estudiantes no saben cómo elegir asignaturas optativas que se ajusten a sus capacidades académicas, aspiraciones profesionales e intereses personales. En esta investigación se recomienda un sistema de recomendación basado en el filtrado de contenidos y en técnicas de filtrado colaborativo para resolver este problema y ayudar a los estudiantes a tomar decisiones informadas sobre las asignaturas optativas.

La estrategia proporciona sugerencias de cursos personalizadas basadas en datos como el rendimiento anterior de los estudiantes, sus áreas de interés específicas y los cursos disponibles. Todos estos datos son examinados por el algoritmo, que recomienda las asignaturas optativas que mejor se adaptan a los objetivos y al nivel de capacidad del estudiante. El objetivo es simplificar el proceso de toma de decisiones voluntarias, reducir la ansiedad ante la decisión y mejorar el rendimiento académico y el compromiso.

La eficacia del concepto se determina mediante una secuencia de pruebas previas y posteriores a la implementación del sistema de recomendación, en las que se evalúan los logros educativos y la satisfacción de los estudiantes. Además, el estudio ofrece una solución in situ y escalable a uno de los mayores problemas de la educación multidisciplinar moderna.

PALABRAS CLAVE: Educación superior, filtrado basado en el contenido, filtrado colaborativo, híbrido, aprendizaje automático.

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